Domain Dropout-Driven Prototype Enhancement Network for Cross-Domain Few-Shot Hyperspectral Change Detection

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***Abstract*—Deep learning (DL) has shown remarkable effectiveness in hyperspectral image change detection (HSI-CD). Cross-domain few-shot learning (CD-FSL) has emerged as an effective solution to label scarcity in the target domain by exploiting labeled source domain data. However, existing methods often suffer from the interference of domain-sensitive channels and the instability of class prototypes. To address these issues, this article proposes a domain dropout-driven prototype enhancement network (DDPEN) for cross-domain few-shot hyperspectral change detection. The method is built upon the metric-based meta-learning to learn transferable knowledge before adapting to new HSI-CD tasks. Specifically, a dual-branch domain dropout-driven feature extractor (DDFE) is constructed to adaptively dropout domain-sensitive channels, thereby extracting domain-invariant features to mitigate domain shift. Subsequently, a history-aware prototype enhancement (HAPE) module is designed to accumulate historical and current prototype information, thus improving the representativeness and stability of class prototypes. Finally, the trained model is transferred to the target domain to construct a change detection inference module under few-shot conditions. Experiments on several hyperspectral datasets illustrate the effectiveness of the proposed algorithm.**

***Index Terms*—Change detection, cross-domain few-shot learning (CD-FSL), domain dropout, hyperspectral images (HSIs).**

I. Introduction

CHARACTERIZED by the high spectral resolution, hyperspectral images (HSIs) provide each pixel with tens to hundreds of nearly contiguous spectral bands [1]. With abundant spectral information, HSI offers significant advantages for fine-grained land cover classification and accurate change detection tasks [2]. HSI change detection (HSI-CD) focuses on identifying and analyzing changes at the same geographic location across different time periods [3]. HSI-CD has attracted increasing attention in recent years and has been widely applied in various fields, such as land cover mapping, agricultural and forestry monitoring, and disaster assessment [4], [5], [6], [7].

In order to make full use of the spectral, spatial, and temporal details of bi-temporal HSIs, numerous methods have been proposed for HSI-CD tasks. Among them, deep learning (DL) methods have demonstrated outstanding capability in extracting abstract and high-level features, especially when trained on large amounts of labeled data [8], [9], [10]. The widely used network architectures in HSI-CD [11] include convolutional neural network (CNN), graph neural network (GNN), and transformer. For example, Wang et al. [12] proposed a 2-D CNN framework that fuses pixel-level and subpixel-level features to improve the performance of HSI-CD. Ou et al. [13] proposed a CNN framework with compact band weighting and multiscale spatial attention to extract the spectral and multiscale spatial features of HSIs. Luo et al. [14] developed a multiscale diff-changed feature fusion network (MSDFFN), which aims to learn the fine-changed features of bi-temporal HSIs. Zhang et al. [15] proposed a multiorder graph convolutional network (MGCN) with channel attention module to repeatedly mix the feature representations of neighborhoods and enhance the difference features of bi-temporal HSIs. Zhao et al. [16] proposed a graph transformer guided multitemporal feature fusion framework to enhance temporal feature interaction. Wang et al. [17] developed the spectral-spatial-temporal transformer (SST-Former) for change detection, focusing on capturing spectral-spatial features from individual HSIs and significant temporal change features between bi-temporal images. Yu et al. [18] proposed a gated spectral-spatial-temporal attention network with spectral similarity filtering to efficiently extract discriminative change features while reducing spectral redundancy. Wu et al. [19] designed a framework that combines information entropy weighting and a bitemporal interactive attention mechanism to enhance the representations of change features. The DL methods mentioned above have demonstrated good performance in HSI-CD across various scenarios. However, these methods typically require a large number of labeled samples for training.

To alleviate the performance degradation caused by label scarcity, a source domain with sufficient annotations is often introduced to assist the target domain learning. However, due

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to the differences in imaging conditions and scene distributions, directly transferring the knowledge from the source domain often leads to suboptimal performance [20], [21]. Therefore, domain adaptation (DA) methods have been proposed to reduce the cross-domain distribution discrepancy and achieve effective knowledge transfer for HSI-CD tasks. For example, Wang et al. [22] proposed a semisupervised domain alignment transformer framework that aligns features from different domains into a shared latent space. Qu et al. [23] proposed an unsupervised feature mutual representation-based graph domain adaptive network (FGDANet) built upon feature mutual representation. The network adopts a pseudo-siamese backbone and a graph DA strategy to align and integrate domain-specific features. Although DA methods have achieved remarkable progress in mitigating the distribution discrepancy between source and target domains, their performance still heavily relies on a certain number of labeled samples or high-quality pseudo labels [24]. To address this issue, researchers have introduced the few-shot learning (FSL) strategy into the DA framework to form the cross-domain FSL (CD-FSL) paradigm, which aims to enhance cross-domain generalization under few-shot conditions [25], [26], [27]. In recent years, several studies have applied the CD-FSL paradigm to HSI-CD tasks.For example, Hou et al. [28] designed a cross-scene FSL network based on intra–inter domain contrastive alignment (CAFSL), effectively integrating contrastive learning with the FSL framework to mitigate the impact of domain shift. Wang et al. [29] proposed a cross-scene FSL framework based on interdomain collaboration (CSIDC-FSL), which integrates and reweights multiscene features via an interdomain joint information module to mitigate cross-scene distribution discrepancy.

The aforementioned CD-FSL methods have achieved promising performance on HSI-CD tasks with limited labeled data. Notably, some CD-FSL methods typically adopt metric-based meta-learning techniques to construct class prototypes and match query samples to these prototypes. However, under few-shot conditions, the class prototypes are typically obtained by averaging randomly selected few support samples during iterative training, which weakens the representativeness and stability of class prototypes. Moreover, some methods attempt to extract domain-invariant features via DA to alleviate domain shift. However, they are often disturbed by channels with cross-domain distribution discrepancy [30], [31]. Such channels introduce domain bias during feature extraction, making the model focus more on domain-specific features than domain-invariant features [32], [33].

To address the above limitations, we propose a domain dropout-driven prototype enhancement network (DDPEN) for cross-domain few-shot hyperspectral change detection. The overall architecture consists of three components: a domain dropout-driven feature extractor (DDFE), a history-aware prototype enhancement (HAPE) module, and a change detection inference module. First, the DDFE module is constructed via a hierarchical multiscale residual (HMR) encoder and a dual-branch domain dropout (DDD) mechanism to extract domain-invariant features from bi-temporal HSIs. The HMR encoder captures the spectral-spatial features of HSIs through multiple multiscale residual convolutional blocks. The DDD mechanism is applied block by block during iterative training to dynamically remove domain-sensitive channels through dual-branch mask-guided domain dropout, thereby enhancing domain-invariant feature extraction and alleviating the domain shift issue. Second, the HAPE module is constructed by combining historical and current prototype information with reverse similarity weighted aggregation and history memory update strategies to improve the stability and representativeness of class prototypes under few-shot conditions. The optimized prototypes are utilized for query matching and model training. Finally, the trained HMR encoder is transferred to construct the change detection inference module, which uses a KNN classifier to produce the change detection result.

The main contributions of this work are summarized as follows.

1) We propose a domain DDPEN based on the CD-FSL paradigm, which can conduct effective knowledge transfer for HSI-CD tasks under few-shot conditions.

2) A domain DDFE is constructed via an HMR encoder and a DDD mechanism, which can effectively extract domain-invariant features to mitigate domain shift.

3) A HAPE module is designed via reverse similarity weighted aggregation and history memory update strategies, which can accumulate historical and current prototype information to enhance the stability and representativeness of class prototypes.

II. Related Work

A. DA and Channel Dropout

DA aims to address the distribution discrepancy between source domain and target domain, known as domain shift [34]. This technique facilitates adaptation between the source and the target domains by aligning their feature distributions [35]. The most typical methods are statistical methods with various metrics that quantify cross-domain distribution discrepancy, such as maximum mean discrepancy [36], Wasserstein distance [37], and graph optimal transport [38]. Qu et al. [39] employed maximum mean discrepancy to achieve multilevel conditional distribution alignment and incorporated cyclic pseudo-label self-training to progressively minimize the distribution discrepancy between domains. Feng et al. [40] proposed a CD-FSL framework based on fractional domain information, which achieves fine-grained cross-domain feature transfer by capturing the intrinsic topological relationships in different domains and aligning them using graph optimal transport. Another well-known kind of method is an adversarial training-based method which is designed to learn domain-invariant features. For example, Zhang et al. [41] developed a generative cross-domain change detection network, which enhances the extraction of domain-invariant features by adversarially training the generator consisting of a morphological encoder and a semantic encoder with the discriminator. However, these methods do not fully explore domain-invariant features, since they pay most attention to designing alignment constraints at the whole feature channels while ignoring the cross-domain distribution discrepancy among various feature channels.

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Some studies have introduced channel dropout strategies to explicitly remove domain-sensitive channels for DA in the field of computer vision. For example, Guo et al. [33] proposed a novel domain-drop framework that employs a domain discriminator (DD) to probabilistically identify domain-sensitive channels. It also incorporates a double consistency constraint to further enhance the extraction of domain-invariant features. Lee et al. [42] designed a random channelwise dropout operation for fully-connected and convolutional layers, which utilize adversarial training via KL divergence loss to extract domain-invariant features from the joint learning of teacher–student network and the DD. Guo et al. [43] proposed random channelwise dropout to progressively drop channels in the randomly selected single layer of ResNet backbone, which utilizes multiple source domains to train a model and then transfer the learned knowledge to the target domain. In addition to being used for DA, some researchers have introduced channel dropout strategies for neural network regularization. For example, Hou and Wang [44] proposed a weighted channel dropout strategy based on the average activation values of the input samples for regularization of deep CNNs. To our knowledge, channel dropout for HSI-CD tasks is less explored. We dig into the implementation of channel dropout for cross-domain few-shot hyperspectral change detection.

B. Cross-Domain FSL

FSL aims to enable models to quickly adapt to new tasks with only a limited number of training samples [45]. In FSL, meta-learning is commonly used as a core strategy, including model-based, metric-based, and optimization-based methods [46], [47], [48], [49]. Among them, metric-based meta-learning methods, especially prototype-based networks, are widely applied in HSI classification. They learn intersample similarities by representing each class with a prototype and matching query samples against these prototypes [50], [51]. Considering the domain shift issue, directly transferring the knowledge learned from source domain to target domain may lead to performance degradation [52]. CD-FSL addresses this problem by integrating FSL with DA techniques, allowing rapid adaptation to target domain tasks under few-shot conditions. Some CD-FSL methods are proposed based on metric-based meta-learning for HSI classification or HSI-CD tasks. For example, Wang et al. [53] introduced a CD-FSL method with classwise attention for HSI classification. Liu et al. [54] proposed multilevel prototype alignment for cross-domain few-shot HSI classification. Wang et al. [55] proposed a cross-heterogeneous domain few-shot learning framework based on bidirectional generation (BiG-FSLF) for HSI-CD. However, these methods typically construct class prototypes by randomly selecting a few support samples and then averaging them in each iteration. Due to the introduction of cross-domain data, there may be significant semantic differences or distribution biases among the support samples of each class. Considering the diversity of support samples, this simple construction approach makes class prototypes susceptible to variations in training episodes and difficult to represent class characteristics. The class prototypes with weakened representativeness and stability degrade the performance of query matching and HSI interpretation tasks.

III. Method

This section describes the proposed DDPEN method, which consists of two processes: cross-domain training and change detection inference. The overall training and testing pipeline is illustrated in Fig. 1.

First, in the cross-domain training stage, the domain DDFE maps all training samples into a learnable metric space, where the Euclidean distance is employed to measure the similarity between support and query samples. This module consists of two components: HMR and DDD. Specifically, HMR jointly models spectral and spatial information through a HMR structure, extracting well-fused spectral-spatial features. On this basis, DDD is introduced to mitigate the interference caused by channels with cross-domain distribution discrepancies. It adopts a blockwise selected dual-branch mask-guided domain dropout mechanism. During training, this strategy dynamically suppresses domain-sensitive channels and encourages the network to focus on domain-invariant features. Subsequently, to obtain more discriminative and stable class prototypes, the HAPE module is introduced. By integrating historical and current prototype information, HAPE performs accumulative optimization of class prototypes across episodes. After completing the entire training process, the model conducts testing and produces final predictions using a KNN classifier.

A. Construction of Source and Target FSL

The training model is constructed based on CD-FSL and meta-learning. It combines source domain data with sufficient annotations and target domain data with few-shot to train the domain DDFE via the HAPE module. In detail, source domain data D_s and target domain data D_t are constructed using the difference images of bi-temporal HSIs from their respective domains. The target domain data D_t are further augmented via multiplicative and additive perturbation strategies; then, we follow the episodic training paradigm for FSL. In each training episode, we randomly select N_s samples for each class to form a support set for each domain, randomly select N_q samples from the remaining samples of each class to form a query set for each domain. We denote the support set of the source domain and the target domain as $S = [S_s, S_t] = \{(x_i^s, y_i^s)\}_{i=1}^{K\times 2N_s}$, K denotes the total class number. $Q = [Q_s, Q_t] = \{(x_j^q, y_j^q)\}_{j=1}^{K\times 2N_s}$ denotes the query set of source domain and target domain. With $K \times 2(N_s + N_q)$ samples in each training episode, the training model aims to extract domain-invariant features.

B. Domain DDFE

1) HMR Encoder: Considering the spectral and spatial details of HSIs, we design a HMR encoder for the cross-domain training process and the change detection inference process. The details of the HMR encoder are illustrated in Fig. 2. Before being fed into the encoder, the input passes through a mapping layer, which performs dimension reduction to reduce computational complexity and unify the spectral

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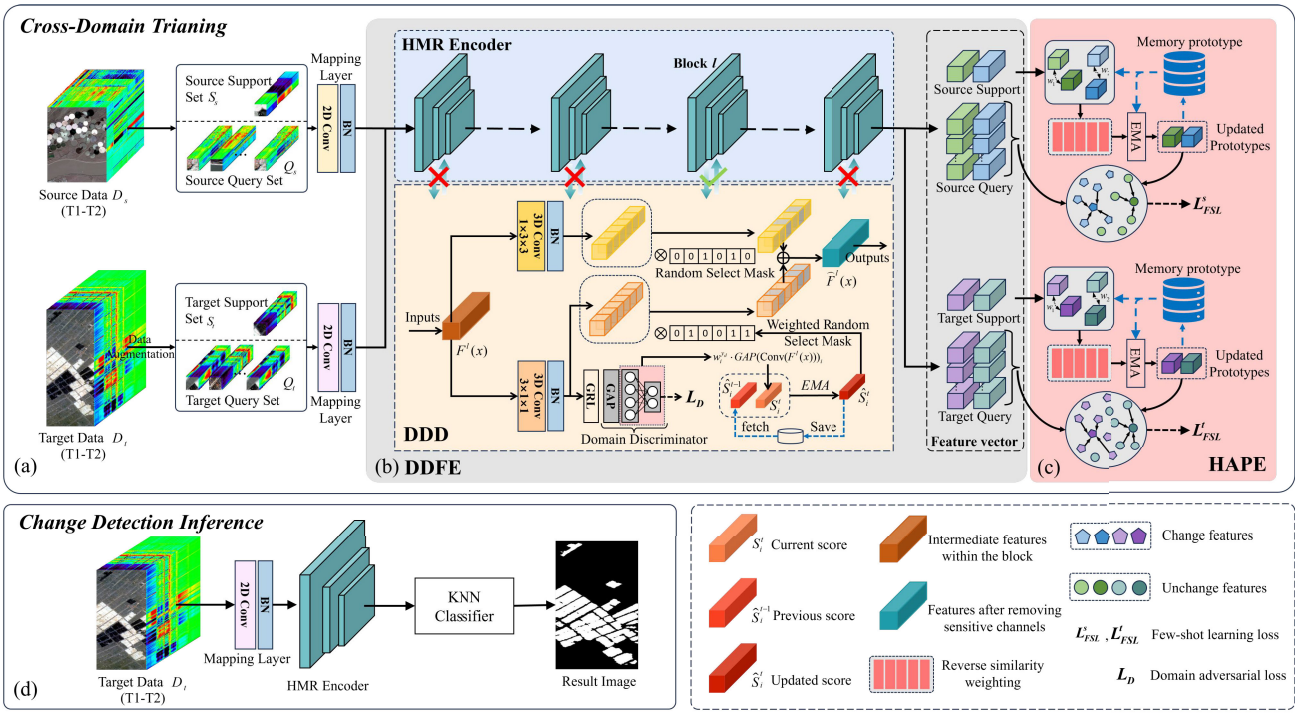


Fig. 1. Overview of the proposed framework. (a) Overall architecture of the cross-domain training model. (b) DDFE module extracts domain-invariant features. In each training iteration, the DDD mechanism is applied block by block to drop domain-sensitive channels. (c) HAPE module enhances class prototype via accumulating historical and current prototype information. (d) Change detection inference module uses the trained HMR encoder and KNN classifier to generate the final change detection map.

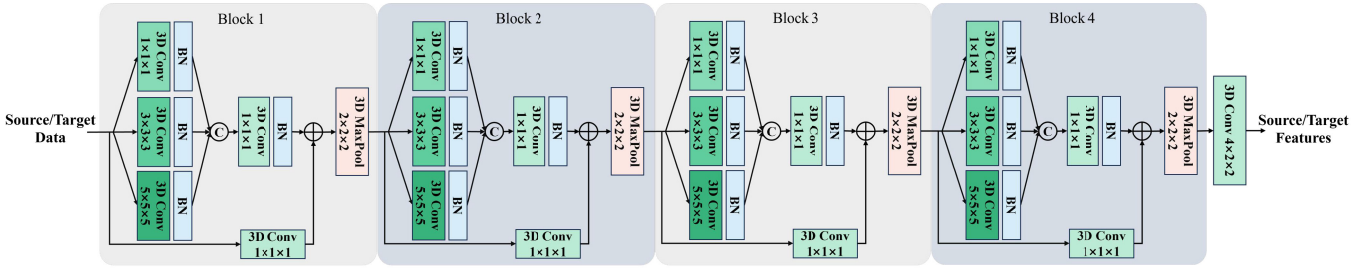


Fig. 2. HMR encoder which consists of four stacked multiscale residual convolutional blocks.

dimensions of the source and target domain data. This step uses a 2-D convolution layer and a batch normalization (BN) layer. The multiscale residual convolution block fuses the output features from 3-D convolution layers with different kernel sizes. The fused features are updated via a 3-D convolution layer and a BN layer. The updated features are added to the output features from the mapping layer and the 3-D convolution layer. Then, a max pool operation is performed to reduce the number of trainable parameters and computational cost. Four multiscale residual convolution blocks are stacked and followed by a 3-D convolution layer to obtain the final spectral-spatial features.

2) Dual-Branch Domain Dropout: As the training model introduces cross-domain data, the domain shift issue exists in the extracted features via the HMR encoder. Motivated by the domain sensitivity differences among feature channels, we design a DDD mechanism to progressively drop domain-sensitive channels in the selected multiscale residual convolution block. The overall framework of DDD consists of blockwise selection and dual-branch channelwise dropout. It is worth noting that the entire DDD mechanism is applied solely to the support set to maintain consistent feature representations and reduce computational overhead.

Blockwise Selection: Different from most previous dropout methods that conduct dropout on a randomly selected layer of the network [33], [42], [43], we propose a domain dropout based on blockwise selection which sequentially selects a single block of the HMR encoder and performs dual-branch channel-wise dropout on its output features during the iterative training process. On the one hand, DDD conducts dual-branch channelwise dropout in a different single block at each iteration, therefore preventing the model from losing too much information in one iteration. On the other hand, DDD conducts dual-branch channelwise dropout in different blocks that output features with different scales, thereby forcing the model to focus on more regions rather than just local regions and to alleviate the overfitting issue in the cross-domain training process.

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Dual-Branch Channelwise Dropout: Instead of treating all channels equally and performing random channelwise dropout [42], [44], we design a dual-branch channelwise dropout mechanism that first generates masks for two parallel branches and then performs mask-guided channelwise dropout. The first branch applies random channelwise dropout, while the second branch leverages a DD and the exponential moving average (EMA) mechanism to identify the domain-sensitive channels, and further introduces the weighted random dropout strategy to suppress the domain-sensitive channels. The dropout is performed on the output feature of the selected block to generate the masked feature vector. The new feature vector is passed through the subsequent blocks to conduct the HAPE module. We denote the input sample as x and its output of the lth block as $F^l \in \mathbb{R}^{C\times H\times W}$, where H, W, C denote the height, width, and channel number.

For the first branch, we conduct a spatial convolution operation to further enhance the local feature representation and apply a low-probability random channel dropout to preserve its structural modeling capability. In detail, we generate a random mask matrix $Mspa^l \in \mathbb{R}^{C\times 1\times 1}$ and drop the channels with randomly sampled indexes $\{r_1, r_2, \cdots r_n\}$ from the C channels of $Mspa^l$, where n is the number of dropped channels. The dropout probability is set to 0.1. For the sampled channel, its corresponding element in $Mspa^l$ is set to 0, and the other elements are set to 1. The random masked output $Fspa^l(x)$ for the lth block is obtained by elementwise multiplication of the convoluted output and random mask matrix

$$\begin{aligned} Mspa^l_{i,1,1} &= \begin{cases} 0, & i \in \{r_1, r_2, \cdots r_n\} \\ 1, & otherwise \end{cases} \\ Fspa^l(x) &= \mathrm{Conv}\left(F^l(x)\right) \odot Mspa^l. \end{aligned} \tag{1}$$

For the second branch, we apply a spectral convolution operation to further refine the feature representations and perform domain-sensitive channel dropout. In detail, we score the domain sensitivity of the channels via DD and drop the channels with a weighted random mask. The DD consists of a global average pooling (GAP) layer, a fully connected layer, and a softmax activation layer, which is used to identify whether the input sample originates from the source domain or the target domain. The channel of the input sample is considered as a domain-sensitive channel if it contributes significantly to binary domain classification, and it is considered as a domain-invariant channel otherwise. For input sample x, let the response of the jth channel after GAP be $GAP(\cdot)_j$. Denote the weight of the fully connected layer corresponding to the true domain label y_d as $w_i^{y_d}$. The domain sensitivity score S_i of the ith channel can be defined as follows:

$$S_i = w_i^{y_d} \cdot GAP\left(\mathrm{Conv}\left(F^l(x)\right)\right)_i,\ i \in \{1, 2, \cdots C\}. \tag{2}$$

In each training episode, only a limited number of samples are used to estimate the domain sensitivity scores of the channels, which often lead to significant score fluctuations across different training episodes. We introduce EMA mechanism to perform exponential smoothing estimation on the domain sensitivity scores of the channels across different training episodes. This operation facilitates more stable score estimation and channelwise dropout. Let S_j^t denote the domain sensitivity score of the ith channel estimated in the tth episode, and denote $\hat{S}_i^{t-1}$ for the previous episode. The domain sensitivity score of the channel is exponentially smoothed as follows:

$$\hat{S}_i^t = \alpha S_i^t + (1-\alpha)\hat{S}_i^{t-1} \tag{3}$$

where α is a smoothing factor, set to a default value of 0.9.

After obtaining stable domain sensitivity scores of the channels $\hat{S}_i^t$, we construct a probability vector $p = \{p_i\}_{i=1}^C$ to quantify the likelihood of each channel being dropped. The probability for the ith channel is defined as follows:

$$p_i = \hat{S}_i^t \Big/ \sum_{i=1}^{C} \hat{S}_i^t \tag{4}$$

where C denotes the total number of channels.

To construct a random mask matrix $Mspe^l \in \mathbb{R}^{C\times 1\times 1}$ for channel dropout, we are inspired by [44] and generate a weight via dropout probability and randomly sampled number for each channel. We then drop m channels with larger weights and set their corresponding elements in $Mspe^l$ to 0. The weighted random masked output $Fspe^l(x)$ for the lth block is obtained by elementwise multiplication of the convoluted output and weighted random mask matrix

$$\begin{aligned} w_i &= \frac{u_i}{p_i} \\ Mspe^l_{i,1,1} &= \begin{cases} 0, & if\ i \in \{r_1, r_2, \cdots r_m\}, \quad w_{r_1} \geq \cdots \geq w_{r_C} \\ 1, & otherwise \end{cases} \\ Fspe^l(x) &= \mathrm{Conv}\left(F^l(x)\right) \odot Mspe^l \end{aligned} \tag{5}$$

where w_i is the weight for channel i, and u_i is a random number sampled from a uniform distribution (0,1). The value of m is discussed in Section IV-E.

With the mask matrices from both two parallel branches, we can obtain the final masked output $\widetilde{F}^l(x)$ for the lth block, which can be represented as follows:

$$\widetilde{F}^l(x) = Fspe^l(x) + Fspa^l(x). \tag{6}$$

C. History-Aware Prototype Enhancement

Following the CD-FSL methods based on metric-based meta-learning, the HAPE module constructs class prototypes using support samples in the feature space and utilizes them to match query samples. To improve the representativeness of the class prototypes, we construct class prototypes by accumulating prototype information across different episodes during training with reverse similarity weighted aggregation and history memory update strategies.

For the support samples $\{x_i^s\}_{i=1}^{2N_s}$ of class c, we first define a memory prototype $P_c^t \in \mathbb{R}^d$ for class c at episode t. Then, we compute an episode-specific temporary prototype E_c^t via a reverse similarity weighted aggregation

$$\begin{aligned} E_c^t &= \frac{\sum_{i=1}^{2N_s} rs_i \cdot \mathrm{DDFE}\left(x_i^s\right)}{\sum_{i=1}^{2N_s} rs_i} \\ rs_i &= 1 - \cos\left(\mathrm{DDFE}(x_i), P_c^{t-1}\right) \end{aligned} \tag{7}$$

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where DDFE(x_i^s) represents the support feature output from the domain DDFE, rs_i represents the reverse similarity between the output of support sample x_i^s and the historical prototype P_c^{t-1}, $\cos(\cdot,\cdot)$ represents cosine similarity.

With (7), HAPE module combines the historical prototype information obtained by the support samples at the previous episode with the support samples at the current episode to construct class prototype. The introduction of additional information may improve the representativeness of the class prototype. When rs_i is equal to 1, (7) degenerates to the equally weighted average operation, while HAPE assigns higher weights to support samples that deviate more from historical prototype, thereby exploiting the intraclass heterogeneity within cross-domain data. To maintain temporal stability and accumulate information across episodes, we apply an EMA update for temporary prototype E_c^t to obtain memory prototype at episode t, which can be represented as follows:

$$P_c^t = \begin{cases} 0, & t = 0 \\ E_c^t, & t = 1 \\ \beta \cdot P_c^{t-1} + (1-\beta) \cdot E_c^t, & t \geq 2 \end{cases} \tag{8}$$

where β is a coefficient controlling the trade-off between historical and current prototype information. In our experiments, β is set to 0.5 to balance historical prototype information with temporary prototype information for more stable cross-episode representation of class prototype.

With the updated prototypes, we can perform matching of query samples to the prototypes. In detail, we first calculate the similarity between each query sample and each class prototype by the Euclidean distance function, then predict the class of the query sample based on the probability that it belongs to different classes. The class of the query sample will be assigned to the class corresponding to the highest probability value. The prediction of the probability that a query sample $x_j^q \in Q$ belongs to class c is calculated as follows:

$$P_\theta\left(y = c | x_j^q \in Q\right) = \frac{\exp\left(-dis\left(\text{DDFE}\left(x_j^q\right), P_c^t\right)\right)}{\sum_{c=1}^{K} \exp\left(-dis\left(\text{DDFE}\left(x_j\right), P_c^t\right)\right)} \tag{9}$$

where Q is the query set, x_j denotes the query sample, and $dis(\cdot)$ denotes a Euclidean distance function.

D. Loss Function

The training model first utilizes the FSL losses generated by matching the query samples with the optimized prototypes, which are calculated as follows:

$$\begin{aligned} L_{\text{FSL}}^s &= E_{S_s,Q_s}\left[\sum_{(x^s,y^s)\in Q_s} -\log P_\theta\left(y^s = c | x^s\right)\right] \\ L_{\text{FSL}}^t &= E_{S_t,Q_t}\left[\sum_{(x^t,y^t)\in Q_t} -\log P_\theta\left(y^t = c | x^t\right)\right] \end{aligned} \tag{10}$$

where L_{FSL}^s and L_{FSL}^t, respectively, denote the FSL loss for source and target domain data.

Besides, the DD loss is introduced to reduce the domain discrepancy between source and target features. Specifically, the DD is embedded in the second branch domain dropout to perform binary domain classification. We adopt a gradient reversal layer (GRL) [30] prior to the DD to enable adversarial DA and learn domain-invariant features. Formally, the DD loss L_D can be represented as follows:

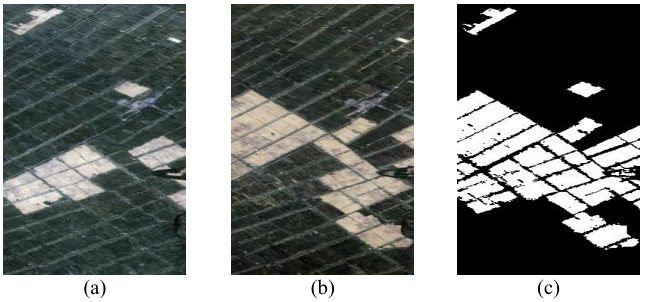
$$\begin{aligned} L_D = &-\frac{1}{K \times 2N_s} \\ &\times \sum_{i=1}^{K\times 2N_s} \Big[y_d^{(i)} \cdot \log\left(DD\left(HMR\left(x_i^s\right)\right)\right) \\ &+ \left(1 - y_d^{(i)}\right) \cdot \log\left(1 - DD\left(HMR\left(x_i^s\right)\right)\right)\Big] \end{aligned} \tag{11}$$

where x_i^s denotes samples drawn from the support set of source domain and target domain. $y_d^{(i)} \in \{0, 1\}$ denotes the true domain label of sample x_i^s, it is set to 0 for samples from source domain and set to 1 for those from target domain. $DD(\cdot)$ represents the DD which can obtain the predicted domain label, $HMR(\cdot)$ represents the HMR encoder.

The final loss combines three components and can be represented as follows:

$$L_{total} = L_{FSL}^s + L_{FSL}^t + L_D. \tag{12}$$

Once the training model is obtained via the total loss, the HMR encoder is transferred to construct the change detection inference module for target domain testing. The testing model utilizes a KNN classifier to complete the binary change detection task and produce the final change detection map.



(a) (b) (c)

Fig. 3. China dataset. (a) Image acquired on May 3, 2006. (b) Image acquired on April 23, 2007. (c) Ground-truth image.

IV. Experiment

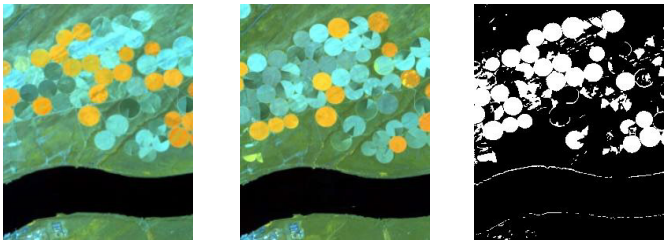
A. Experimental Dataset

In this work, several hyperspectral datasets [56], [57] are applied to evaluate the performance of the proposed method.

1) China Dataset: This dataset comprises two HSIs captured near Yancheng, Jiangsu, China, on May 3, 2006, and April 23, 2007, respectively, as shown in Fig. 3(a) and (b). Acquired by the Hyperion sensor onboard the EO-1 satellite, the images, mainly covering agricultural areas, have a spatial size of 420 × 140 pixels with 155 spectral bands. It includes 18 277 labeled changed pixels and 40 467 unchanged pixels.

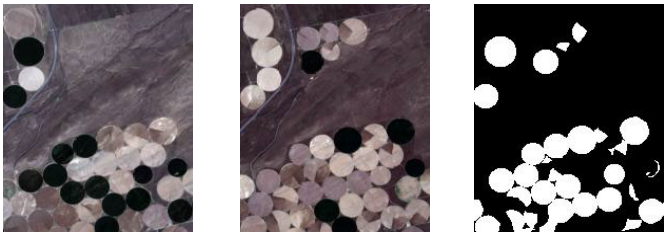
2) USA Dataset: This dataset consists of two HSIs collected over Hermiston, OR, USA, on May 1, 2004, and May 8, 2007, respectively, as shown in Fig. 4(a) and (b). The images were acquired using the Hyperion sensor onboard the EO-1

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(a) (b) (c)

Fig. 4. USA dataset. (a) Image acquired on May 1, 2004. (b) Image acquired on May 8, 2007. (c) Ground-truth image.



(a) (b) (c)

Fig. 5. Benton dataset. (a) Image acquired on May 1, 2004. (b) Image acquired on May 8, 2006. (c) Ground-truth image.

satellite, covering irrigated farmland areas. Each image has a spatial resolution of 307 × 241 pixels and contains 156 spectral bands. There are 16 727 labeled changed pixels and 57 260 unchanged pixels.

3) Benton Dataset: This dataset contains two HSIs acquired over Benton, OR, USA, on May 1, 2004, and May 8, 2007, respectively, as shown in Fig. 5(a) and (b). Captured by the EO-1 Hyperion sensor, the images mainly depict irrigated agricultural regions, with a spatial size of 180×225 pixels and 159 spectral bands. The dataset includes 9921 labeled changed pixels and 30 579 unchanged pixels. To comprehensively evaluate the cross-domain generalization ability of the proposed method, we design three cross-domain experiments based on three hyperspectral change detection datasets. Specifically, we construct two cross-domain tasks by using USA as the source domain and China and Benton as the target domains. In addition, we establish a third cross-domain task by using China as the source domain and USA as the target domain.

B. Experimental Setup

We choose several metrics to quantitatively evaluate model performance, including overall accuracy (OA), Kappa coefficient, precision, $F1$-score, as well as training and testing times. To validate the effectiveness of the proposed method, we compare it against several recently published state-of-the-art approaches: MSDFFN [14], AIWSEN [19], MGCN [15], CMGA [11], SST-Former [17], DAN [58], FGDANet [23], BiG-FSLF [55], and CSIDC-FSL [29]. To evaluate the performance of the proposed method, we adopt the following experimental configuration. For the target domain, five labeled samples are randomly selected from both the changed and unchanged classes. For the source domain, 500 labeled samples per class are randomly selected to provide transferable knowledge for cross-domain change detection. In the DDPEN model, the input patch size is set to 9×9, and the unified feature dimension after the mapping layer for both the source and target domains is fixed at 128. For the second branch domain dropout, the dropout rate of domain-sensitive channels is set to 0.16 based on the parameter analysis. The network is trained using the SGD optimizer with a momentum of 0.9 and a weight decay of 0.0005. The initial learning rate is set to 0.01 and decayed by a factor of 0.1 after 80% of the total 2000 training epochs. An early stopping strategy with a patience of 20 evaluation intervals is employed to prevent overfitting. In each training episode, the support set contains one labeled sample per class, while the query set contains 19 labeled samples per class. All methods are implemented in PyTorch, and the number of labeled samples is kept consistent across experiments to ensure a fair comparison. All experiments are conducted on an NVIDIA GeForce RTX 3090 GPU.

TABLE I
RESULTS OF ABLATION STUDY ON DIFFERENT MODULES

	Method	HAPE	DDD	OA	Kappa	Training time(s)
China	Base	×	×	94.88	88.41	123.98
	+HAPE	✓	×	95.90	90.49	142.73
	+DDD	×	✓	96.34	91.47	181.2
	DDPEN	✓	✓	**96.68**	**92.26**	201.08
Benton	Base	×	×	92.86	81.44	105.91
	+HAPE	✓	×	94.64	85.67	110.87
	+DDD	×	✓	94.18	84.28	194.85
	DDPEN	✓	✓	**95.60**	**86.60**	209.81
USA	Base	×	×	92.02	75.95	158.84
	+HAPE	✓	×	93.30	80.60	161.33
	+DDD	×	✓	93.77	81.25	226.75
	DDPEN	✓	✓	**94.80**	**84.78**	238.75

C. Ablation Study

1) Ablation Study on Different Modules: To validate the effectiveness of DDD and HAPE modules, we perform ablation experiments on three datasets. We first establish a baseline model consisting solely of the HMR encoder and a standard FSL framework. Subsequently, the DDD and HAPE modules are, respectively, added to the baseline to quantify their individual impact.

As shown in Table I, both the proposed HAPE and DDD modules make positive contributions to change detection performance. With the integration of the HAPE module, the OA on China, Benton, and USA datasets increases by 1.02%, 1.78%, and 1.28%, respectively. Meanwhile, the corresponding Kappa values rise by 2.08%, 4.23%, and 4.65%. These improvements demonstrate that HAPE effectively enhances the representativeness of class prototypes. Similarly, incorporating the DDD module leads to consistent improvements in both OA and Kappa across all datasets. Specifically, OA increases by 1.46%, 1.75%, and 1.32%, while Kappa improves by 3.06%, 2.84%, and 5.30%, respectively. These results confirm the effectiveness of the proposed domain dropout mechanism. Compared with using either DDD or HAPE alone, integrating

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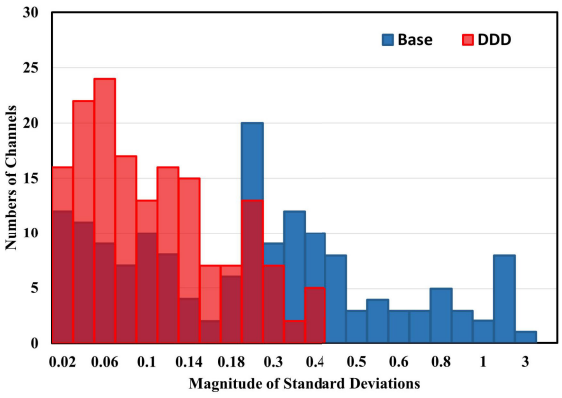


Fig. 6. Channel-wise domain robustness analysis. The blue bars represent the results of the baseline (base) model, which consists solely of a HMR encoder combined with a standard FSL strategy. The red bars represent the addition of the DDD modules to the baseline model.

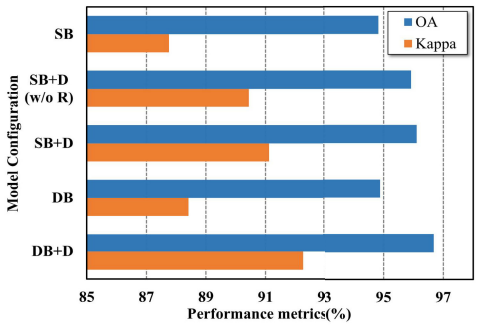


Fig. 7. Ablation study on the dual-branch structure and channel dropout strategy. SB: single branch, SB + D: add channel dropout strategy to single branch, w/o R: without using the weighted random dropout strategy, DB: dual branch without channel dropout strategy, DB + D: the complete DDD module.

both modules consistently achieves the highest OA and Kappa values.

2) Channelwise Domain Robustness Analysis: To further validate the effectiveness of the DDD module, we analyze the channelwise domain robustness of the feature maps produced by the last convolutional layer of the HMR encoder.

We use China as the source domain and the USA as the target domain. Specifically, GAP is applied to each channel of the feature maps to obtain channel responses. The average response of each channel is then computed separately for the source and target domains. For each channel, the standard deviation between the source domain and target domain average responses is calculated to measure channelwise domain robustness. A smaller standard deviation represents more consistent channel responses across domains, indicating that the corresponding channel is less affected by domain shift and thus exhibits stronger domain robustness.

As shown in Fig. 6, after introducing the DDD module, channelwise standard deviations are more concentrated at smaller values. In contrast, the baseline model exhibits a more dispersed distribution of channelwise standard deviations. This result suggests that the DDD module effectively suppresses domain-sensitive channels rather than equally treating all channels. Therefore, DDD encourages the model to rely more on domain-invariant features, which are consistent with the design objective of the DDD module and contributes to improved cross-domain generalization.

3) Ablation Study on the Dual-Branch Structure and Dropout Strategy: To further analyze the components of the DDD module, we design several ablation settings as follows.

1) *SB:* The first branch of DDD is removed, and only the second branch of DDD is retained. The channel dropout strategy and the weighted random dropout strategy are removed, only the DD is used for adversarial learning.

2) *SB* $+D$ $(w/o\ R)$: Based on (1), the channel dropout strategy is introduced. The mask is generated only based on the discriminator scores, without using the weighted random dropout strategy.

3) *SB* $+D$: Based on (2), the weighted random dropout strategy is combined with the discriminator scores to generate the final mask.

4) *DB:* Both branches of DDD are retained, but the channel dropout strategy is not applied.

5) *DB* $+D$: The complete DDD module.

As shown in Fig. 7, both the dual-branch structure and the channel dropout strategy effectively improve model performance. Introducing weighted random dropout further enhances the effect of the feature mask, thereby improving OA and Kappa value. Compared to a single branch, the complete DDD module exhibits superior stability and discriminative ability.

D. Change Detection Results

Table II presents the change detection performance of DDPEN and the comparison methods. The results show that DDPEN achieved the best performance across most indicators. As shown in Figs. 8–10, the visualization results further confirm the significant advantage of DDPEN.

1) Experiment Results of China Dataset: Due to the limited availability of labeled samples, the model distinguishing change and nonchange regions in the farmland dataset faces challenges. As shown in Fig. 8, compared to five DL-based approaches, three FSL-based methods better suppress false positives. Specifically, SST-Former, MSDFFN, MGCN, and AIWSEN produce false positive detections in the upper-left region. In addition, DAN boundary recognition is unclear, and the image shows thin, elongated noise. AIWSEN and CMGA cannot accurately identify the boundaries of change regions. FGDANet produces jagged and discontinuous boundaries. BiG-FSLF and CSIDC-FSL exhibit incomplete delineation of small-scale change areas. In contrast, DDPEN achieves more accurate segmentation of fine-scale change structures, preserving boundary continuity.

In addition, as shown in Table II, DDPEN achieves the best results across all major performance metrics. Compared with CSIDC-FSL, DDPEN improves performance by 2.15% in OA, 4.11% in Kappa, 2.29% in precision, 1.80% in recall, and 2.05% in $F1$-score. Notably, AIWSEN achieves an abnormally high recall of 99.70%. This is because, with limited training samples, the model tends to classify most pixels as "change" to

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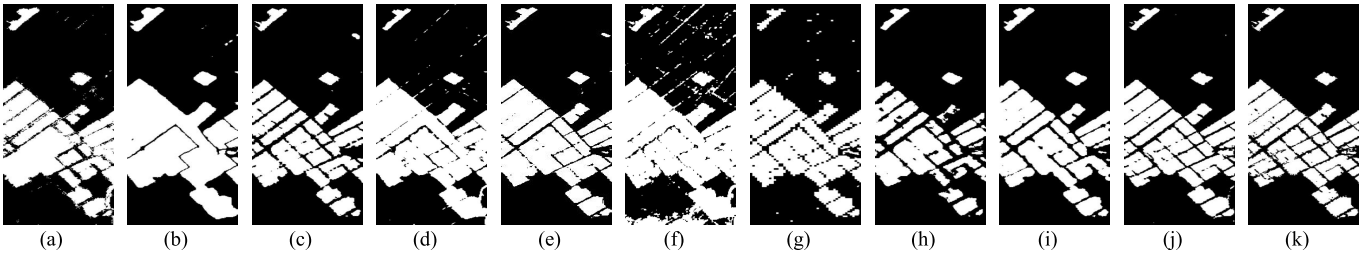


Fig. 8. Change detection maps of different methods on China dataset. (a) MSDFFN. (b) AIWSEN. (c) MGCN. (d) CMGA. (e) SST-Former. (f) DAN. (g) FGDANet. (h) BiG-FSLF. (i) CSIDC-FSL. (j) DDPEN. (k) Ground-truth image.

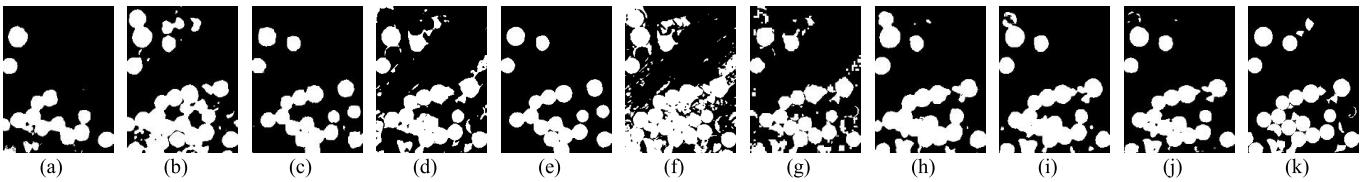


Fig. 9. Change detection maps of different methods on Benton dataset. (a) MSDFFN. (b) AIWSEN. (c) MGCN. (d) CMGA. (e) SST-Former. (f) DAN. (g) FGDANet. (h) BiG-FSLF. (i) CSIDC-FSL. (j) DDPEN. (k) Ground-truth image.

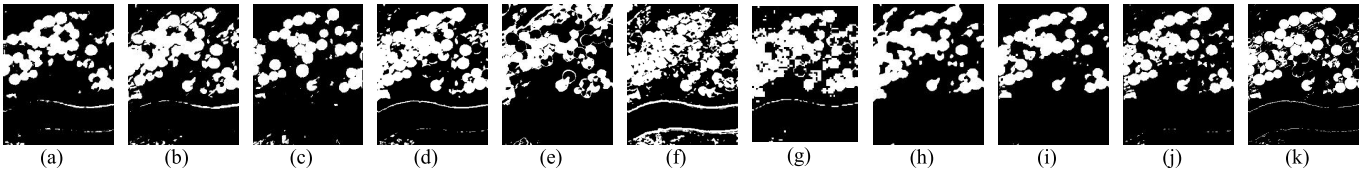


Fig. 10. Change detection maps of different methods on USA dataset. (a) MSDFFN. (b) AIWSEN. (c) MGCN. (d) CMGA. (e) SST-Former. (f) DAN. (g) FGDANet. (h) BiG-FSLF. (i) CSIDC-FSL. (j) DDPEN. (k) Ground-truth image.

TABLE II
PERFORMANCE OF CD ON THREE DATASETS

Dataset	Metric	MSDFFN	AIWSEN	MGCN	CMGA	SST-Former	DAN	FGDANet	BiG-FSLF	CSIDC-FSL	DDPEN
China	OA	91.53	91.10	94.69	92.98	93.38	87.06	94.10	92.67	94.53	**96.68**
	Kappa	80.76	80.71	87.48	84.46	84.89	72.87	86.49	82.65	88.15	**92.26**
	F1	87.02	87.44	93.74	92.20	89.77	86.19	90.88	88.87	94.08	**96.13**
	Precision	83.10	77.87	94.17	90.81	86.43	85.27	87.84	91.32	93.72	**96.01**
	Recall	91.33	**99.70**	93.55	94.54	93.37	90.55	94.03	87.18	94.46	96.26
Benton	OA	88.37	84.27	91.41	92.58	91.67	82.90	93.30	93.42	93.24	**95.60**
	Kappa	64.97	63.20	75.51	80.94	76.19	62.07	82.37	82.57	83.67	**86.60**
	F1	72.01	73.62	87.73	92.75	81.51	80.49	86.85	88.42	91.63	**93.30**
	Precision	87.78	63.64	90.25	93.25	89.41	79.06	83.63	86.79	90.60	**93.76**
	Recall	61.05	87.32	85.82	92.58	74.89	87.81	90.34	90.62	92.82	**92.87**
USA	OA	89.20	88.77	88.11	91.15	82.49	77.28	89.05	93.10	93.37	**94.80**
	Kappa	68.27	70.53	64.79	77.43	53.39	46.66	72.14	80.08	80.24	**84.78**
	F1	75.59	77.95	82.39	91.57	64.91	72.70	79.37	86.26	90.12	**92.39**
	Precision	77.06	69.93	83.65	93.25	59.19	71.43	68.94	86.26	91.92	**93.35**
	Recall	74.18	88.04	81.34	91.15	71.86	78.45	**93.51**	90.31	88.60	91.52

avoid missing true changes. As a result, many unchanged pixels are incorrectly labeled as changed, which increases false positives and lowers OA.

2) Experiment Results of Benton Dataset: In this dataset, several methods exhibit omissions in detecting circular-like targets. As shown in Fig. 9, MSDFFN and SST-Former miss some change regions, such as the lower-left area of the image. DAN and CMGA exhibit irregular white patches and fine noise. FGDANet produces coarse and inaccurate boundaries. AIWSEN, BiG-FSLF, and CSIDC-FSL mistakenly identify unchanged areas as circular change regions, such as the circular patterns in the upper-left area. In contrast, DDPEN demonstrates stronger ability in delineating change region boundaries and effectively suppresses false positives in the upper-left area. It also shows fewer missed detections in change regions.

As shown in Table II, M2DFSL attains the best results among all evaluation metrics. Compared with CSIDC-FSL, 2.36% in OA, 2.93% in Kappa, 3.16% in precision, 0.05% in recall, and 1.67% in $F1$-score.

3) Experiment Results of USA Dataset: The USA dataset contains complex land-cover changes, making linear regions in the lower part challenging for all methods. As shown in Fig. 10, the overall image of DAN appears fragmented with

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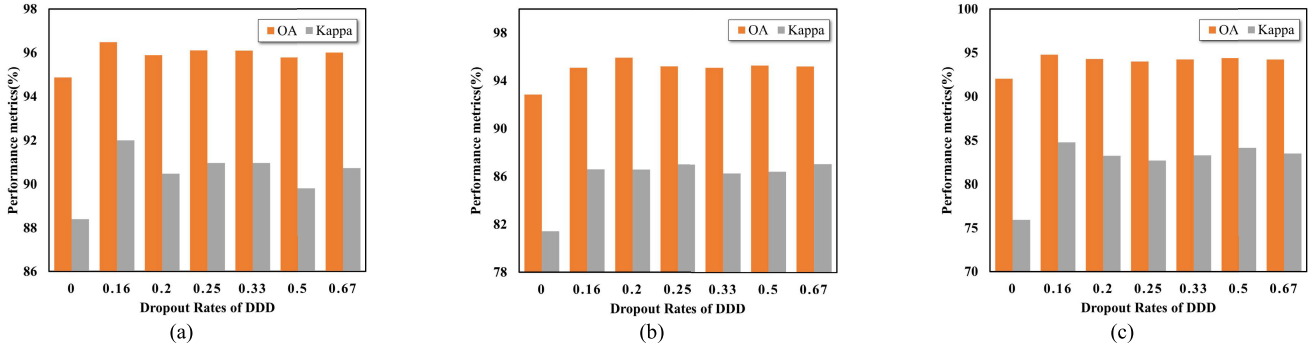


Fig. 11. Performance of different dropout rates in the second branch domain dropout on three datasets. (a) China. (b) Benton. (c) USA.

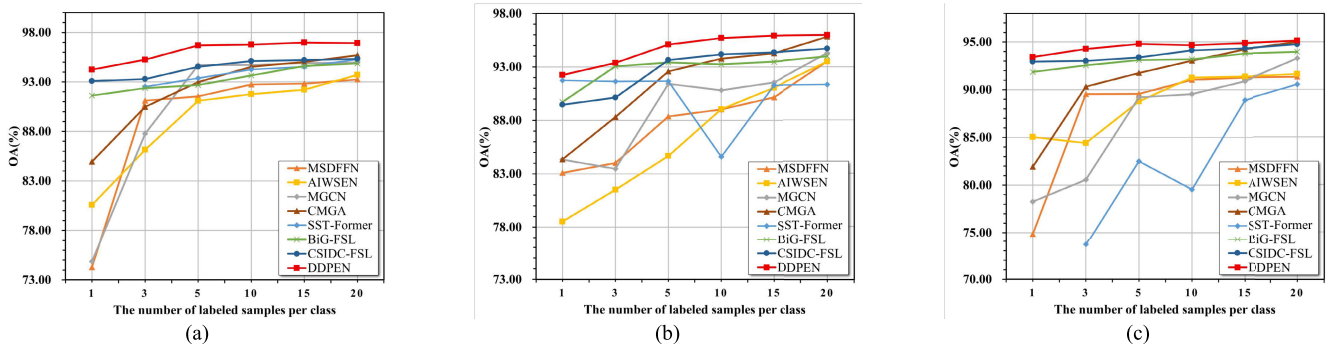


Fig. 12. OA of different methods with different training samples on three datasets. (a) China. (b) Benton. (c) USA.

unclear boundaries. SST-Former fails to clearly distinguish between changed and unchanged regions in the upper part of the image. MSDFFN and MGCN miss circular change regions, such as the circular area in the center of the image. AIWSEN, FGDANet, and BiG-FSLF fail to clearly delineate the boundaries of circular target change regions, causing them to appear merged into a single area. CSIDC-FSL misses some small-scale change regions. In contrast, DDPEN performs better in detecting small-scale changes and provides clearer delineation of their boundaries.

Moreover, as shown in Table II, DDPEN achieves the highest performance across all major evaluation metrics. Compared with CSIDC-FSL, DDPEN improves performance by 1.43% in OA, 4.54% in Kappa, 1.43% in precision, 2.92% in recall, and 2.27% in $F1$-score. Notably, SVM achieves an extremely high precision of 99.99% but a relatively low recall of 51.43%. This indicates that limited training samples cause the model to conservatively predict the "change" class, leading to many true change pixels being missed.

E. Parameter Analysis

This study systematically analyzes the impact of dropout rate and training sample size on change detection performance.

1) Effect of Dropout Rates: To investigate the effect of dropout rates on change detection performance, we followed the settings in [33] and selected six typical dropout rates {0.16, 0.20, 0.25, 0.33, 0.50, 0.67}. These were applied to the second branch within the DDD module, covering a spectrum from mild to severe dropout intensities. This range enables a systematic evaluation of how different dropout strengths influence hyperspectral feature learning. As shown in Fig. 11, applying appropriate dropout significantly improves overall performance across three datasets compared to the baseline model without dropout. On average, OA increases by approximately 2%–4%, and Kappa improves by about 3%–8%. It is worth noting that the optimal dropout rate varies across datasets, which may be attributed to differences in spectral dimensions and class distributions. Considering both performance gains and experimental reproducibility, we ultimately selected 0.16 as the default dropout rate for subsequent experiments.

2) Effect of Training Samples: We further evaluated the impact of varying the number of labeled training samples per class in the target domain. The quantity was set to {1, 3, 5, 10, 15, 20} to cover scenarios ranging from extreme low-shot (one-shot) to relatively sufficient few-shot (20-shot) settings. As shown in Figs. 12 and 13, the proposed method was compared with SST-Former, MSDFFN, MGCN, AIWSEN, CMGA, BiG-FSLF, and CSIDC-FSL, evaluated by OA and the Kappa coefficient. Across all shot settings and datasets, our method consistently achieved the highest performance. Notably, in the one-shot setting, the proposed method achieved OA values of 94.24%, 92.24%, and 93.42% on China, Benton, and USA datasets, respectively. The corresponding Kappa coefficients were 86.76%, 79.84%, and 81.57%. These results outperform all competing methods under the same conditions. These experimental results fully demonstrate the effectiveness and robustness of our approach in few-sample environments. They indicate that our algorithm

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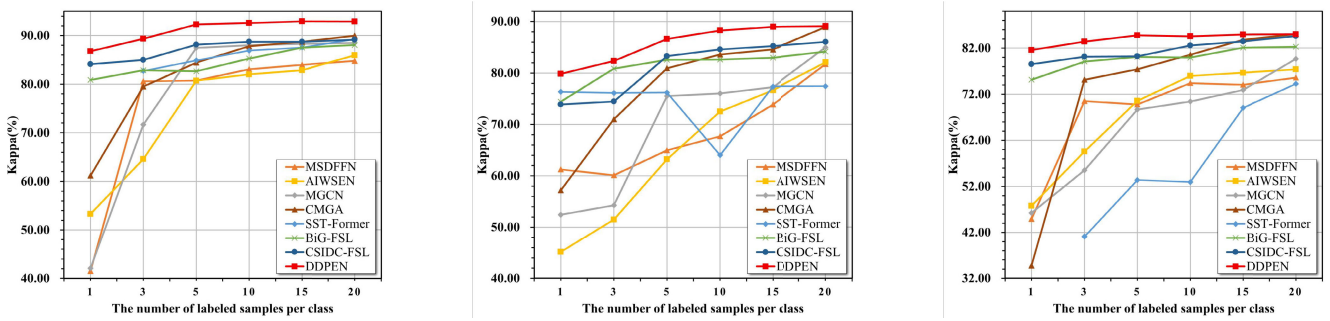


Fig. 13. Kappa of different methods with different training samples on three datasets. (a) China. (b) Benton. (c) USA.

TABLE III
PARAMETERS, FLOPS, AND COMPUTATIONAL TIME WITH DIFFERENT METHODS ON THREE DATASETS

Dataset	Metric	MSDFFN	AIWSEN	MGCN	CMGA	SST-Former	DAN	FGDANet	BiG-FSLF	CSIDC-FSL	DDPEN
China	Training time(s)	24.64	26.27	15.29	3.49	36.34	8.41	38.16	106.26	236.10	201.08
	Testing time(s)	0.17	0.14	0.03	0.05	5.26	1.22	0.02	4.36	4.65	5.34
	Params(M)	26.46	12.23	0.09	0.11	2.50	0.35	2.02	0.34	0.45	0.58
	FLOPs(G)	9.78	0.58	2063.01	-	1.61	0.17	141.60	0.42	0.43	0.95
Benton	Training time(s)	25.11	28.31	101.23	1.9	37.78	7.86	30.44	88.39	159.49	209.81
	Testing time(s)	0.15	0.16	0.03	0.03	3.66	0.83	0.01	3.16	3.15	3.77
	Params(M)	26.46	12.23	0.09	0.11	2.50	0.35	2.02	0.34	0.45	0.58
	FLOPs(G)	9.78	0.58	4596.17	-	1.61	0.17	141.60	0.42	0.43	0.95
USA	Training time(s)	27.68	31.93	38.11	5.53	37.01	8.43	38.16	123.87	211.30	238.75
	Testing time(s)	0.15	0.16	0.07	0.08	6.64	1.51	0.02	5.52	5.86	6.47
	Params(M)	26.46	12.23	0.10	0.11	2.51	0.35	2.02	0.34	0.45	0.58
	FLOPs(G)	9.78	0.58	993.17	-	1.62	0.17	113.97	0.42	0.43	0.95

adapts well to various small-sample scenarios and excels at HSI-CD.

V. DISCUSSION

A. Complexity Analysis

We compared the running times, FLOPs, and parameters of different algorithms. The specific details are shown in Table III. From the perspective of model complexity, the parameter scale of DDPEN is slightly higher than that of BiG-FSLF and CSIDC-FSL, yet it remains significantly lower than most mainstream comparison methods. This indicates that the proposed model enhances feature representation capability without introducing excessive redundant parameters. For methods such as MGCN and FGDANet, the higher FLOPs mainly result from graph construction and graph-based relational modeling. In contrast, the FLOPs of DDPEN remain at a moderate level, achieving a reasonable trade-off between representational capacity and computational cost.

In terms of training efficiency, the training time of DDPEN is higher than that of SST-Former, MSDFFN, and BiG-FSLF. This is primarily because the proposed method requires a relatively larger number of training epochs to enable the DDD module to sufficiently identify and analyze domain-sensitive information across different scales and channels. In addition, the three few-shot methods exhibit relatively longer inference times during testing due to their reliance on KNN-based inference. As evidenced by Table I, our framework built upon BiG-FSLF repeatedly performs KNN over both the training and testing sets, leading to redundant forward computations and increased inference cost. Specifically, the prediction process requires additional feature matching and similarity computation, which introduces extra computational steps beyond a direct forward pass and consequently increases the overall inference overhead.

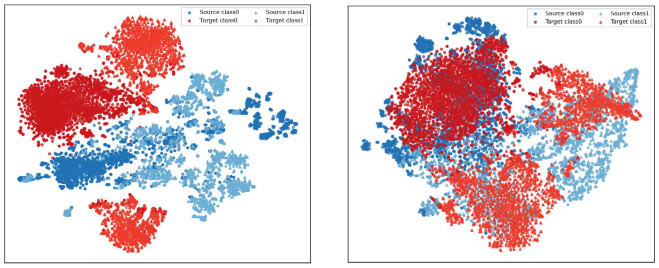


Fig. 14. Visualization result images by T-SNE. (a) Before. (b) After.

B. Feature Visualization

In this t-SNE visualization, the USA dataset serves as the source domain and the China dataset as the target domain. Red points correspond to the source domain and blue points correspond to the target domain, while different color intensities within the same domain indicate different classes. Fig. 14(a) illustrates the feature distribution before applying DDPEN. It can be observed that the source and target domains are clearly separated in the embedding space, with a large distance between their distribution centers and limited overlap,

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TABLE IV
CROSS-DOMAIN GENERALIZATION VERIFICATION ON SANTA AND RIVER DATASETS

Target domain	Method	OA	Kappa	F1
Santa	BiG-FSLF	92.98	85.14	89.96
	CSIDC-FSL	93.74	86.88	93.44
	DDPEN	**93.97**	**87.34**	**93.67**
River	BiG-FSLF	91.61	55.01	71.85
	CSIDC-FSL	91.41	57.11	78.42
	DDPEN	**93.65**	**62.84**	**81.41**

indicating a noticeable cross-domain distribution shift. In contrast, Fig. 14(b) presents the T-SNE visualization after introducing the DDPEN module. In the feature space, the alignment between source and target samples is significantly improved, and the cross-domain overlap increases. Features of the same class from different domains also show a more consistent distribution. This demonstrates the effectiveness of the proposed DDPEN method, which helps reduce distribution differences caused by domain-specific features and enables the model to learn more domain-invariant and discriminative representations.

C. Cross-Domain Generalization Verification

To further validate the generalization capability of the proposed DDPEN model, we additionally conduct a domain generalization experiment using the USA dataset as the source domain while using the Santa and River datasets [58] as the target domains. The USA dataset was collected by the EO-1 Hyperion sensor and mainly contains mountainous and farmland areas. The Santa dataset was acquired by the AVIRIS sensor and mainly covers mountainous and farmland regions, whereas the River dataset was collected by the EO-1.

Hyperion sensor and primarily focuses on river regions. The USA-Santa cross-domain datasets reflect the cross-land-cover scenario, while the USA-River cross-domain datasets reflect the cross-sensor scenario. As shown in Table IV, our method maintains competitive performance in both the cross-sensor scenario (USA → Santa) and the cross-land-cover scenario (USA → River) compared with other CDFSL methods. These findings suggest that the proposed method can effectively handle the domain shifts in the cross-sensor and cross-land-cover scenarios, and exhibits good generalization capability.

VI. CONCLUSION

In this article, we propose a DDPEN for cross-domain few-shot HSI-CD. The method introduces a DDD mechanism into the HMR encoder to achieve effective hyperspectral feature extraction. During feature extraction, the DDD mechanism adaptively identifies and drops domain-sensitive channels that exhibit large cross-domain distribution discrepancies. This process mitigates the adverse effects of domain shift. Besides, we design a HAPE module that leverages both historical and current prototype information to enhance the feature representations of class prototypes. Experimental results on several benchmark datasets demonstrate that the proposed method achieves superior detection performance while relying on fewer labeled samples.

However, the DDPEN model requires retraining when encountering a new target domain, which limits its practical applicability. Besides, the HAPE module employs a fixed coefficient to control the trade-off between historical and current prototype information, which may not result in the best effect of prototype enhancement. Future work could extend the current single-source domain setting to a multisource scenario, thereby enabling the model to adapt to diverse target domains without repeated retraining. In addition, future work could explore adaptive parameter learning strategies to further improve the effectiveness of prototype enhancement.

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